

Discovering efficient gate sets for fault-tolerant quantum computation

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We introduce SWIFTBOT, a pipeline for discovering efficient gate sets for fault-tolerant quantum computation. It operates in two modes. In *universal* mode, it ranks $(\mathcal{C}, g)$ pairs — a finite base group $\mathcal{C} \subset \text{SU}(d)$ extended by a single gate g — by the Quantum Circuit Overhead Q_T , the asymptotic cost of approximating Haar-uniform targets. In *application-specific* mode, it minimizes the mean synthesis depth over a structured target family picked out by the application. A third stage matches surviving pairs against a curated catalog of transversal QEC codes, magic-state distillation/cultivation/code-switching protocols, and certified exact-synthesis rings, flagging which pairs admit an end-to-end fault-tolerant stack. At $d = 3$ the universal mode yields the first systematic Q_T ranking across three Σ -series subgroups, with arbitrary-SU(3) extensions on the largest approaching the Kesten–McKay floor within 32%, while structured diagonal and Howard–Vala extensions sit 3–70 $\times$ above. In application-specific mode, a non-canonical rational-phase extension beats canonical Clifford+ T by 6.8 $\times$ on a lattice-gauge target family, showing that the best general-purpose and best target-specific extensions can differ by nearly an order of magnitude. The pipeline, registered groups, fault-tolerance catalog, and reproduction data are released with the paper.

I. INTRODUCTION

Fault-tolerant quantum computation requires a finite universal gate set whose members have cheap physical realizations. The tension between these requirements is sharp: the Eastin–Knill theorem [1] forbids any non-trivial quantum error-correcting code from admitting a universal *transversal* gate set. Practical designs therefore split the gate set into a transversal subgroup $\mathcal{C} \subset \text{SU}(d)$ that the error-correcting code protects cheaply, plus one or more non-transversal extension gates g made fault-tolerant by any of several routes: magic-state distillation [2, 3], magic-state cultivation [4], code switching [5], or beyond-stabilizer codes with native non-Clifford transversal action [6–8]. Each extension gate g comes with its own cost: a dedicated magic state, a code that supports it transversally, a distillation or cultivation circuit, or some combination of the above.

Why qudits. The qubit version of this story has a standard resolution: Clifford+ T [2, 9], with Clifford transversal on Steane, Reed–Muller, and Bravyi–Haah families and T distilled via triorthogonal MSD [2, 3]; single-qubit synthesis into this gate set is solved with certified T -count bounds [10, 11]. This toolkit is mature but not closed: as we show for the binary icosahedral base $2I$ (Section IV), systematic $(\mathcal{C}, g)$ exploration beyond the canonical choice remains productive even at $d = 2$, and four developments motivate extending the search into qudit registers at $d > 2$.

Qudit hardware has matured on several fronts. Trapped-ion platforms have demonstrated single-qudit registers up to $d = 7$ [12] and operate multi-ion chains at lower d ; superconducting cavities encode $\gtrsim 10$ Fock levels

per mode with coherence times approaching the best qubit counterparts [13]; reconfigurable neutral-atom arrays provide individually addressable sites at the hundred-qudit scale [14]; and solid-state spin- S systems offer $d = 2S + 1$ native encodings [15, 16]. The binary-encoding overhead qubits impose on inherently higher- d physics is a cost, not a feature of these architectures.

Such hardware development has been matched by advances in theoretical proposals for fault tolerance, both on the QEC side and on the non-transversal-gate side.

Qudit codes have progressed beyond the qubit toolkit. Gottesman’s prime- d stabilizer construction [17] spelled out the foundational split from the qubit case; beyond stabilizer, the *nonadditive* Kubischta $((7, 2, 3))_{2I}$ code [6] and Denys–Leverrier *covariant* $2T/2O$ codes [7, 8] carry transversal non-Clifford groups that sidestep the Anderson–Jochym–O’Connor qubit-stabilizer no-go [18]. HEP-motivated bosonic proposals such as the $2T$ -quocit encode specific discrete subgroups natively at $d = 8$ [19]. Non-local qLDPC memories like the bivariate-bicycle code [20] need connectivity graphs whose natural hardware substrates — the same platforms above — make qudit qLDPC codes [21, 22] a convergence rather than two separate programs.

Distillation and cultivation are advancing in parallel. Qutrit triorthogonal distillation [23, 24] reaches $\gamma \approx 1.51$, beating qubit triorthogonal codes below ~ 300 physical qudits; asymptotically-good qudit constructions [25, 26] deliver constant-overhead MSD. Magic-state cultivation [4, 27, 28] is currently qubit-native and surface-code-specific, and an open qudit analogue would close one of the two remaining arguments for $d = 2$ canonicalization (Section VII.1).

Applications drive the practical case for qudits. Lattice-gauge simulations of $\text{SU}(N)$ live on $|\mathcal{C}|$ -dimensional per-link registers that fit qudit architectures natively $(|\Sigma(36 \times 3)| = 108 \text{ states per link, encoded on three qutrits})$

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plus two qubits [29, 30]); experimental trapped-ion realizations of $U(1)$ and $SU(2)$ gauge dynamics already use native qudits [31, 32], and native-qudit QEC constructions are in active development for $\mathbb{Z}_N$ [33] and $SU(2)$ [34] lattice gauge theories. Beyond lattice gauge, higher- d registers reduce the Trotter cost of continuous-variable bosonic simulations, cut the ancilla overhead of fermion-orbital encodings in chemistry and materials science, and absorb large-spin degrees of freedom (magnetism, $S > \frac{1}{2}$ impurities) without binary unpacking [15, 16].

qLDPC codes and qudit extensions. A complementary insight: modern codes like qLDPC may be better understood as qudit codes than as collections of qubit codes. The gate set accessible on a $[[n, k, d]]$ qLDPC code like the $[[144, 12, 12]]$ gross code [20] through fold-transversal Cliffords and code automorphisms is *not* the product group $\text{Clifford}(2)^{\otimes k}$: recent work on bivariate-bicycle codes shows the fold-transversal Cliffords act collectively on the logical qubits [35], with the general code-automorphism-to-logical-gate construction [36] and the remaining-logical-Cliffords syndrome-extraction cost [37] now spelled out. Built from code-structured Pauli-string operators, the accessible set is more naturally read as a specific finite subgroup $\mathcal{C} \subset SU(2^k)$ — a single-qudit object at $d = 2^k$ rather than a tensor product of qubit gates. The corresponding non-Clifford completion could then be a single unitary g on the k -qubit logical space: a high-weight entangling gate in the qubit basis, but a single-qudit extension at $d = 2^k$. Distilling the one $|\psi_g\rangle$ magic state is structurally distinct from distilling k separate $|T\rangle$'s, and potentially cheaper: constant-rate qudit distillation [25, 26] makes this a live possibility rather than a formal rephrasing. The $(\mathcal{C}, g)$ ranking this paper formalizes applies once $\mathcal{C}$ is concretely identified and the relevant t -design irreps at $d = 2^k$ are tractable; a gross-code case study is left to future work.

No comparable consensus exists at $d > 2$. A three-parameter family $HV(z, \gamma, \varepsilon)$ of qudit T -gates at prime d is known [38], but no systematic comparison of members has been done. The finite-subgroup classification of $SU(d)$ is complete for $d \leq 4$ [39–43] but open beyond, and transversal-gate classifications for stabilizer codes are available for qubit stabilizer codes [18]; qudit analogues exist for specific code families but not in the same exhaustive form. Prime- d distillation protocols exist [23], qudit synthesis for Shor-class algorithms has been studied [44], and Ross–Selinger-style exact synthesis has recently been extended to Clifford+ R at $d = 3$ [45]. What is missing is a quantitative ranking: given a base group $\mathcal{C}$ and an extension g , how efficiently does $(\mathcal{C}, g)$ synthesize target unitaries?

Slowik, Dulián, and Sawicki [46] supplied the right figure of merit. The *Quantum Circuit Overhead* $Q_T(\mathcal{C}, g)$ is the asymptotic exponent in the word length of an ε -accurate approximation: the number of gates scales as $(1/\varepsilon)^{Q_T}$, bounded below by the Kesten–McKay optimum Q_{opt} and saturated by the “golden gates” of Sarnak [47].

This paper uses Q_T to rank candidates. We introduce

SWIFTBOT, an agentic workflow that enumerates $(\mathcal{C}, g)$ pairs — in principle for any dimension d , tested in the present paper at $d \leq 4$ where classification bounds are known — screens each for universality via Sawicki’s criterion, evaluates Q_T numerically [46], and cross-references surviving pairs against a curated catalog of 12 transversal QEC codes (stabilizer and beyond-stabilizer), 10 magic-state distillation / cultivation / code-switching protocols spanning five target-gate families, and two certified exact-synthesis rings.

We make three contributions. First, at $d = 2$ we reproduce the Slowik *et al.* numerics at the percent level (Section IV). Second, we establish the first systematic Q_T ranking at $d = 3$ across the three irreducible-adjoint Σ -series subgroups with $|\mathcal{C}| \in \{216, 648, 1080\}$, using Howard–Vala, rational-phase, and arbitrary- $SU(3)$ extensions (Section V). Third, applied to a gate set relevant for lattice-gauge theory simulation [19, 29, 48], a non-canonical rational-phase extension beats Clifford+ T by $6.8\times$ in mean synthesis depth (Section VI). Together these show that the best general-purpose extension and the best target-specific extension can differ by nearly an order of magnitude—a distinction invisible when Q_T alone is reported.

Code and data are released with this paper [49]. Section II reviews the QCO framework, Section III sketches the pipeline, Sections IV to VI present the results, and Section VII closes with a fault-tolerance audit and open problems.

II. BACKGROUND

We briefly review the three ingredients used throughout: the Quantum Circuit Overhead Q_T [46] as our primary figure of merit, the Sawicki criterion [50] for ruling out non-universal extensions without running QCO, and the Howard–Vala family [38] and discrete-subgroup-approximation (DSA) primitive-gate series [29, 48] as the principal sources of candidate extensions at $d \geq 3$.

II.1. δ -approximate t -designs and Q_T

The Quantum Circuit Overhead Q_T is a property of a *generating set* $\mathcal{S} = \mathcal{C} \cup \{g, g^{-1}\}$, not of the base group $\mathcal{C}$ alone; a poor choice of extension g can make the Cayley random walk converge slowly on $SU(d)$ even on a base group with a large spectral gap. We therefore never quote Q_T without specifying the extension.

Let $\mathcal{S} = \mathcal{C} \cup \mathcal{C}^{-1} \cup \{g, g^{-1}\} \subset SU(d)$ be a symmetric generating set consisting of a finite subgroup $\mathcal{C}$ and a single extension g . Consider the t -design operator on the t -th tensor power of the defining representation,

$$T_t(\mathcal{S}) = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \pi_t(s), \quad \pi_t = \bigoplus_{\lambda \in \Lambda_t} \pi_\lambda^{\oplus m_\lambda}, \quad (1)$$

TABLE I. Kesten–McKay floor $Q_{\text{opt}} = \log |\mathcal{C}| / \log(1/\delta_{\text{opt}})$ for the finite subgroups referenced in this paper, with $\delta_{\text{opt}} = 2\sqrt{|\mathcal{C}| - 1}/|\mathcal{C}|$.

d Group	$ \mathcal{C} $	Q_{opt}
2 hurwitz (projective A_4)	12	4.19
2 $\text{Clifford}_1 / 2T$	24	3.47
2 $2O$ (binary octahedral)	48	3.09
2 BI (binary icosahedral $2I$)	120	2.81
3 $\Sigma(60) \simeq A_5$	60	3.00
3 $\Sigma(36 \times 3)$ (Valentiner)	108	2.83
3 $\Sigma(72 \times 3)$	216	2.69
3 $\Sigma(216 \times 3)$ (= qutrit Clifford_1)	648	2.54
3 $\Sigma(360 \times 3)$	1080	2.49

where Λ_t is the set of $\text{SU}(d)$ highest weights that appear in the decomposition of $(\mathbb{C}^d)^{\otimes t} \otimes (\mathbb{C}^d)^{* \otimes t}$ and π_λ is the associated irreducible representation. The t -*design spectral radius* is

$$\delta_t(\mathcal{S}) = \max_{\lambda \in \Lambda_t \setminus \{0\}} \|T_t(\mathcal{S})\|_{\pi_\lambda}, \quad (2)$$

the largest spectral norm across non-trivial irreps. A smaller δ_t means the random walk generated by $\mathcal{S}$ equidistributes faster on $\text{SU}(d)$.

For an ε -net of $\text{SU}(d)$ in the operator norm, Słowiak *et al.* [46] show that the number of Solovay–Kitaev-style words required scales as $(1/\varepsilon)^{Q_T}$, with

$$Q_T(\mathcal{S}) = \frac{\log |\mathcal{C}|}{\log(\delta_t^{-1}(\mathcal{S}))}, \quad (3)$$

in the limit $t \rightarrow \infty$. The Kesten–McKay bound [51–53] for a k -regular Cayley graph gives, in the $|\mathcal{C}| \rightarrow \infty$ asymptotic we use throughout,

$$\delta_t \geq \delta_{\text{opt}} = \frac{2\sqrt{|\mathcal{C}| - 1}}{|\mathcal{C}|}, \quad (4)$$

and therefore

$$Q_T \geq Q_{\text{opt}} = \frac{\log |\mathcal{C}|}{\log(1/\delta_{\text{opt}})}, \quad (5)$$

with equality for “golden gates” [47]. Table I lists Q_{opt} for every base group in our registry that we use as a baseline below.

In practice Q_T is estimated at finite t by sampling the ensemble in Eq. (1) with a Monte-Carlo extension gate g (the *rnd* baseline) or a deterministic one, then evaluating Eq. (2) via sparse spectral decomposition on each irrep of π_t . The feasibility ceiling at $d = 3$ is set by representation dimensions [46]: the largest irrep at $t = 50$ has dimension $\sim 10^5$, whereas $t = 15$ peaks at $\sim 4 \times 10^3$ and is tractable. At $d = 2$, the Słowiak *et al.* $Q_T(t)$ convergence curves [46] reach stabilization at the percent level by $t \gtrsim 30$. At $d = 3$ our reference point is weaker: we observe the Tier-1 ($t = 5$) and Tier-2 ($t = 10$) δ values agree to four decimals

for all structured extensions (Section V), which strongly suggests stabilization, but without an independent $t \rightarrow \infty$ reference we cannot tightly bound the residual Q_T error at $d = 3$. An asymptotic extrapolation $Q_T(t \rightarrow \infty)$ from a $Q_T(t)$ sweep is a natural next step. We quote Q_{opt} alongside Q_T throughout to expose the gap to optimality.

II.2. Pre-screening with the Sawicki criterion

Not every candidate extension needs a QCO run. Sawicki and Karnas [50] proved that $\mathcal{S}$ generates a dense subgroup of $\text{SU}(d)$ if and only if

- (i) the adjoint representation of $\mathcal{S}$ is irreducible, and
- (ii) $\langle \mathcal{S} \rangle$ — the subgroup generated by $\mathcal{S}$ — is not contained in any finite subgroup of $\text{SU}(d)$ (checked against the Blichfeldt classification caps).

The Sawicki–Karnas criterion is stated on $\mathcal{S}$, but the irreducibility of $\text{Ad}_{\mathcal{S}}$ has a convenient sufficient check on $\mathcal{C}$ alone. Since $\mathcal{C} \subset \mathcal{S}$, every matrix commuting with Ad_s for all $s \in \mathcal{S}$ also commutes with Ad_c for all $c \in \mathcal{C}$, so $\text{Comm}(\text{Ad}_{\mathcal{S}}) \subseteq \text{Comm}(\text{Ad}_{\mathcal{C}})$; if $\dim \text{Comm}(\text{Ad}_{\mathcal{C}}) = 1$ already, then $\dim \text{Comm}(\text{Ad}_{\mathcal{S}}) = 1$ for every g . Condition (i) is therefore *automatic* whenever $\text{Ad}_{\mathcal{C}}$ is irreducible. Schur’s orthogonality relation reduces the check on $\mathcal{C}$ to an $\mathcal{O}(d^2)$ -memory character sum,

$$\dim \text{Comm}(\text{Ad}_{\mathcal{C}}) = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} |\chi_{\text{Ad}}(c)|^2, \quad (6)$$

where $\chi_{\text{Ad}}(c) = \text{tr Ad}_c$. When $\dim \text{Comm}(\text{Ad}_{\mathcal{C}}) > 1$, condition (i) on $\mathcal{S}$ can still hold if g mixes $\mathcal{C}$ ’s Ad -invariant subspaces; in this case we re-check on the full generating set $\mathcal{C} \cup \{g, g^{-1}\}$ via the joint kernel of the $(\text{Ad}_s \otimes I - I \otimes \text{Ad}_s^T)$ stack, valid on non-closed sets. Bases with reducible $\text{Ad}_{\mathcal{C}}$ (e.g. $\Sigma(36 \times 3)$, $\Sigma(60) \simeq A_5$, the $s_{60 \times 4}$ family) are not automatically excluded; we defer them in this paper for compute reasons (Section V) but flag them as natural targets for a follow-up. Condition (ii) is enforced at finite cost by running a bounded closure $\langle \mathcal{S} \rangle \cap \text{Cap}(n)$ against the classification of finite subgroups of $\text{SU}(d)$ (orders ≤ 120 at $d = 2$, ≤ 1080 at $d = 3$, ≤ 5040 at $d = 4$) [39–43]; we use cap sizes $n = 120, 1080, 5040$ respectively. An extension is flagged **finite** if closure terminates below the cap, **universal_likely** if closure survives the cap and condition (i) holds, and **infinite_but_not_dense** if condition (i) fails but closure is open.

II.3. Candidate extensions

At $d = 2$ the canonical extension is $T = P(\pi/4) = \text{diag}(1, e^{i\pi/4})$, giving the Clifford+ T set with certified synthesis via Ross–Selinger [10] and Kliuchnikov–Maslov–Mosca [11]. We also consider diagonal $P(\theta)$ for $\theta \in$

$\{\pi/8, 2\pi/9, \pi/9, \pi/18\}$, the Parzanchevski–Sarnak supergolden gates [54], and *arbitrary-SU(d)* extensions — unconstrained single-qudit unitaries sampled uniformly from the Haar measure on $SU(d)$, labeled `rnd` in the tables below.

At prime d the Howard–Vala family [38] generalizes T via three integers $(z, \gamma, \varepsilon) \in \mathbb{Z}_d^3$. At $d = 3$ — the only dimension used in our survey — the $\det = 1$ condition requires a global $\omega^{1/3}$ twist, so the diagonal entries live on 9th (not 3rd) roots of unity [38, Eq. 27]:

$$\text{HV}(z, \gamma, \varepsilon) = \mathcal{N} \cdot \text{diag}(1, \zeta_9^{v_1}, \zeta_9^{v_2}), \quad (7)$$

with $\zeta_9 = e^{2\pi i/9}$, $v_1 = (6z + 2\gamma + 3\varepsilon) \bmod 9$, $v_2 = (6z + \gamma + 6\varepsilon) \bmod 9$, and $\mathcal{N}$ a phase fixing $\det \text{HV}(z, \gamma, \varepsilon) = 1$. (For primes $p > 3$ the base root is ζ_p and v_k is the cubic-in- k recurrence solution $v_k = (k/12)[\gamma + k(6z + (2k-3)\gamma)] + k\varepsilon \pmod{p}$ [38, Eq. 24].) Our panel uses $(z, \gamma, \varepsilon) \in \{(1, 1, 0), (1, 2, 0), (2, 1, 0)\}$: the first is Campbell’s magic gate [23], the second is the Howard–Vala Eq. (27) variant (Clifford-equivalent to the first at $d = 3$ [38] — hence identical δ in Table III), and the third is a non-Clifford-equivalent orbit. The full group $\{U_v\}$ at $d = 3$ factorizes as $\mathbb{Z}_9 \times \mathbb{Z}_3$ modulo phases [38, Eq. 35]; these three samples cover two of the Clifford-equivalence orbits.

In addition to the Haar-uniform extensions above, discrete-subgroup approximations of lattice gauge simulations [29, 48, 55] pick out a *structured* set of target unitaries that the chosen extension must synthesize cheaply. For each qudit group, the continuous degree of freedom reduces to a trace rotation $U_{\text{Tr}}(\theta)$ that decomposes into a finite set of $R_Z(\alpha_c \theta)$ rotations whose coefficients α_c depend on the specific subgroup and encoding. This motivates a *target-family* figure of merit: the mean breadth-first-search depth to synthesize the required R_Z rotations within $\varepsilon \leq 10^{-2}$. We instantiate it for one specific subgroup in Section VI.

III. SWIFTBOT PIPELINE

SWIFTBOT is a five-stage agentic pipeline built around the QCO kernel of [46]. Numerical stages are deterministic; an LLM supervisor intervenes only at stage boundaries to rank candidates, propose extensions, and decide when to stop. All headline numbers below come from deterministic panels that bypass the LLM; the agent is used for exploratory search not reported here. Full code and a command-line driver are released with the paper [49].

Stage 1 (group registration) computes projective closures $\langle \mathcal{G} \rangle \subset SU(d)/Z(SU(d))$ from seed unitaries and stores each group under a content-addressable key. The registry covers the canonical $d \in \{2, 3, 4\}$ subgroups used here: Pauli–Clifford, the projective $A_4 \subset \text{PU}(2)$ (registered as `hurwitz`), and $2I$ (BI) at $d = 2$; the $\Sigma(m \times 3)$ series for $m \in \{36, 72, 216, 360\}$ at $d = 3$; and selected Fairbairn subgroups at $d = 4$.

Stage 2 (universality screen) evaluates $\dim \text{Comm}(\text{Ad}_{\mathcal{C}})$ via Eq. (6) and runs a bounded closure against the classification caps $\{120, 1080, 5040\}$ [39–43]. Pairs whose closure terminates below cap are tagged `finite` and skipped; reducible-adjoint pairs are tagged diagnostically.

Stage 3 (QCO numerics) invokes the kernel of [46] as a subprocess. Two optimizations help: (i) every gate has the form $gcg^\dagger$ with $c \in \mathcal{C}$, so $\pi_\lambda(gcg^\dagger) = \pi_\lambda(g)\pi_\lambda(c)\pi_\lambda(g)^\dagger$ replaces $|\mathcal{C}|$ exponentials with one; (ii) the $\pi_\lambda(c)$ lists are persisted to disk keyed by a fingerprint of $\mathcal{C}$ and the highest weight λ . Together these give $3\text{--}6\times$ speedup at $d = 3$. Results are stored as per-sample `QTRecords` in a SQLite cache with full provenance.

Stages 4–5 (fault-tolerance) match each surviving $(\mathcal{C}, g)$ against a curated catalog of QEC codes (Steane, Reed–Muller, Bravyi–Haah, Kubischta 2I, qutrit triorthogonal, ternary Golay, Pollatsek–Ruskai PI, Kubischta–Teixeira PI, and more) and distillation protocols (Bravyi–Kitaev, Bravyi–Haah, Campbell–Anwar–Browne, programmable MSD, Ouyang–Jing–Brennen (OBJ) code-switching). The catalog is curated for tractability, not exhaustive; Hastings–Haah sublogarithmic-overhead codes [56] and the full Litinski resource landscape [57] are notable out-of-scope omissions that would be natural additions. Unmatched pairs return a *research-needed* marker with an EC Zoo [58] pointer rather than guessing.

IV. VALIDATION AT $d = 2$

We validate the SWIFTBOT QCO kernel against the $d = 2$ reference dataset of Slowik, Dulián, and Sawicki [46] on two qubit bases: the Clifford group ($|\mathcal{C}| = 24$) and the 12-element projective $A_4 \subset \text{PU}(2)$ (registered `hurwitz`), each paired with $P(\pi/4)$ and with arbitrary-SU(2) extensions. We also run a systematic 10^4 -sample search on the binary icosahedral group $2I$ (BI, $|\mathcal{C}| = 120$), which carries the Kubischta nonadditive transversal code [6]. All runs are at $t = 50$, matching the reference. A zeroth-order check—replaying the δ -parser on all 5050 rows of the reference file and recomputing Q_T —matches the reference to $\mathcal{O}(10^{-15})$, isolating any later disagreement to the spectral kernel. Headline results are summarized in Table II: Clifford+ $P(\pi/4)$ yields $Q_T = 51.9$ (reference ≈ 52); arbitrary-SU(2) extensions agree at the few-percent level.

Two remarks. First, Clifford+ T has Q_T roughly $14\times$ larger than the arbitrary-SU(2) baseline (51.9 vs 3.79)—i.e. its approximation-exponent overhead is worse by an order of magnitude; commensurability between $P(\pi/4)$ and the Clifford orbit slows the random walk [46]. This is a Q_T claim, not a verdict on Clifford+ T itself: when the target is structured (Section VI), the ranking inverts. Second, the BI ratio $Q_T/Q_{\text{opt}} \approx 1.27$ at 10^4 samples is far closer to saturation than Clifford’s and arrives with a cataloged fault-tolerant code [6]; the empirical δ distribution on BI has median 0.38 ($Q_T \simeq 4.97$) and 5th-percentile 0.31, and a 2000-resample bootstrap on the best-of- 10^4 Q_T gives a tight 95% CI of [3.57, 3.69]. The best-of- 10^4 floor

TABLE II. $d = 2$ validation of the SWIFTBOT QCO kernel against the reference dataset of Slowik, Dulián, and Sawicki [46], all at $t = 50$. The Clifford+ $P(\pi/4)$ row is deterministic; rnd rows report the best Q_T across 20 Haar-SU(2) samples for Clifford and hurwitz, and 10^4 samples for BI. Q_{opt} is the Kesten–McKay floor of Eq. (5); BI has no published QCO reference.

Base ($ \mathcal{C} $)	Ext.	Q_T		Q_{opt}
		this work	[46]	
Clifford (24)	$P(\pi/4)$	51.91	≈ 52	3.47
Clifford (24)	rnd	3.79	≈ 3.7	3.47
hurwitz (12)	rnd	4.34	≈ 4.0	4.19
BI (120)	rnd	3.57	—	2.81

is the natural super-golden candidate predicted by [47].

V. $d = 3$ SURVEY

With the SWIFTBOT kernel validated at $d = 2$ (Section IV), we turn to the first systematic Q_T ranking at $d = 3$. The survey covers the three irreducible-adjoint Σ -series subgroups of SU(3)— $\Sigma(72 \times 3)$, $\Sigma(216 \times 3)$, and $\Sigma(360 \times 3)$ —each extended by a common panel of eight candidate gates. (We use the $\Sigma(m \times 3)$ notation of [40, 43], where $|\Sigma(m \times 3)| = 3m$; the three bases thus have orders 216, 648, 1080.) The adjoints are irreducible — the 8-dimensional $\mathfrak{su}(3)$ representation does not split — by the character tables of [43]. The smaller groups $\Sigma(36 \times 3)$ (the 108-element crystallographic Valentiner analogue) and $\Sigma(60) \simeq A_5$ both have reducible adjoint at $d = 3$: the 8-dimensional $\mathfrak{su}(3)$ representation splits into smaller irreps of the subgroup, as can be read directly from the character tables [43] or verified via Eq. (6) with $\dim \text{Comm}(\text{Ad}_C) > 1$. These are not automatically non-universal: the Σ -invariant subspaces of $\mathfrak{su}(3)$ are not aligned with the Cartan–root decomposition (the Fourier element $V \in \Sigma(36 \times 3)$ mixes them), so even a generic diagonal extension lifts $\dim \text{Comm}(\text{Ad}_S)$ to 1 and the Sawicki criterion is satisfied on $\mathcal{S}$. We defer $\Sigma(36 \times 3)$ and A_5 from this paper purely for compute reasons — the Tier-1/Tier-2 budget was set by the three irreducible-adjoint Σ -series bases — and flag a $\Sigma(36 \times 3)/A_5$ panel as natural follow-up work.¹

V.1. Panel specification

The panel fixes eight extensions per base group, with identical parameters across the three groups. Three are

Howard–Vala variants HV(1, 1, 0), HV(1, 2, 0), HV(2, 1, 0) (see Eq. (7)). Three are diagonal phase gates $P(\theta)$ at angles $\theta \in \{2\pi/9, \pi/9, 2\pi/5\}$, chosen to span both the cyclotomic-9th-root sector relevant to Section VI and the coarser $2\pi/5$ angle that generates no low-order closure against any of the three base groups. The remaining two slots per group are independent arbitrary-SU(3) batches of size 10, drawn freshly per subprocess from the Haar measure on SU(3); these serve as the *rnd* baseline of [46]. All eight extensions survive the Stage 2 universality screen of Section II.2: each Σ -series base has one-dimensional adjoint commutant by Eq. (6), and the bounded closures stay open past the $|\mathcal{C}| \leq 1080$ cap of [39–43].

The panel is evaluated in two passes. An initial Tier-1 screen at $t = 5$ — the largest design order at which the full 3×8 panel completes in ~ 17 min on a single 32-core workstation — fixes the ranking and feeds Sections V.3 and V.4 below. For the headline values of Table III and Fig. 1 we re-run the panel at $t = 10$ (Tier-2); all structured-extension δ agree bit-for-bit with $t = 5$ except HV(2, 1, 0) on the two smaller groups, which lifts off its finite- t floor at $t = 10$ (see item (i) below). Fixed extensions use `sample_size = 1` (the gate is deterministic); rnd batches use `sample_size = 10`. Each (group, extension) pair is dispatched as an independent `qco` subprocess with `OMP_NUM_THREADS=1` and `--workers 8` to keep the per-irrep spectral evaluation of Eq. (2) single-threaded while exploiting thread-level parallelism across the $|\mathcal{C}|$ matrix actions described in Section III. All numerical values reported below are reproduced bit-identically from the SQLite caches released with the paper [49].

V.2. Survey ranking

Table III lists every (group, extension) pair at $t = 10$, ranked within each group by δ . The $Q_T / \log |\mathcal{C}|$ column divides out the base-group scale and exposes the spectral-gap content of the extension.

Fig. 1 elaborates three findings against Table III.

Finding 1 (structured-extension degeneracy). Five of six structured extensions give δ identical to four decimals across all three base groups, bit-for-bit the same as at $t = 5$. HV(2, 1, 0) is the only exception: it lifts from the $t = 5$ floor $\delta = 0.5000$ on the smaller groups to $\delta \approx 0.58$ at $t = 10$, while $\Sigma(360 \times 3)$ stays at 0.5698. The degeneracy is structural, not a finite- t artifact.

Finding 2 (linear Q_T growth with $\log |\mathcal{C}|$). Because δ is essentially constant, Q_T grows linearly with $\log |\mathcal{C}|$ for the degenerate extensions: a larger base group needs proportionally more depth to densely cover SU(3), so the normalized $Q_T / \log |\mathcal{C}|$ (rightmost column) collapses the three groups onto the same value.

Finding 3 (arbitrary-SU(3) gains with $|\mathcal{C}|$). Best-sample rnd δ drops from 0.26 to 0.18 from $\Sigma(72 \times 3)$ to $\Sigma(360 \times 3)$; an arbitrary-SU(3) draw couples to every Lie-algebra direction including the subdominant channels that grow with $|\mathcal{C}|$. Statistically significant rnd distributions are

¹ TODO: insert $\Sigma(36 \times 3) + \{\text{panel}\}$ and $A_5 + \{\text{panel}\}$ Q_T data when compute is available; the SWIFTbot Stage-2 screen now classifies $\Sigma(36 \times 3) + \text{rnd}$ as `universal_likely` via the corrected $\text{Comm}(\text{Ad}_S)$ check on generators.

TABLE III. $d = 3$ survey at $t = 10$. Each fixed extension is deterministic (δ from a single evaluation); the rnd row per group reports the best of two independent Haar draws, each itself a single-sample evaluation at this design order. Kesten–McKay floor Q_{opt} : 2.692, 2.544, 2.495 for $\Sigma(72, 216, 360 \times 3)$. Rows are ranked within each group by δ ; all structured extensions match their $t = 5$ values to four decimals except HV(2, 1, 0) on the two smaller groups, where the $t = 5$ floor at $\delta = 0.5000$ lifts at $t = 10$.

Group	Extension	kind	δ	Q_T	$Q_T/\log \mathcal{C} $
$\Sigma(72 \times 3)$	rnd (best of 2)	rnd	0.2605	3.996	0.744
	HV(2, 1, 0)	howard_vala	0.5927	10.277	1.911
	$P(2\pi/5)$	angle	0.6545	12.681	2.360
	HV(1, 1, 0)	howard_vala	0.6764	13.750	2.559
	HV(1, 2, 0)	howard_vala	0.6764	13.750	2.559
	$P(2\pi/9)$	angle	0.8830	43.208	8.041
	$P(\pi/9)$	angle	0.9698	175.561	32.671
$\Sigma(216 \times 3)$	rnd (best of 2)	rnd	0.2107	4.157	0.643
	HV(2, 1, 0)	howard_vala	0.5833	12.011	1.856
	$P(2\pi/5)$	angle	0.6545	15.273	2.360
	HV(1, 1, 0)	howard_vala	0.6764	16.561	2.559
	HV(1, 2, 0)	howard_vala	0.6764	16.561	2.559
	$P(2\pi/9)$	angle	0.8830	52.039	8.041
	$P(\pi/9)$	angle	0.9698	211.443	32.671
$\Sigma(360 \times 3)$	rnd (best of 2)	rnd	0.1784	4.052	0.579
	HV(2, 1, 0)	howard_vala	0.5698	12.419	1.775
	$P(2\pi/5)$	angle	0.6545	16.478	2.360
	HV(1, 1, 0)	howard_vala	0.6764	17.867	2.559
	HV(1, 2, 0)	howard_vala	0.6764	17.867	2.559
	$P(2\pi/9)$	angle	0.8830	56.145	8.041
	$P(\pi/9)$	angle	0.9698	228.127	32.671

deferred to Sections V.3 and V.4.

V.3. Apples-to-apples Haar panel

To separate the base-group effect from rnd-extension variance, we generated a single batch of ten arbitrary-SU(3) extensions (`seed=42`) and evaluated the panel identically on all three groups. Table IV lists the best-sample δ per matrix.

Across the $N = 100$ Haar panel (same `seed = 42`; the first ten matrices coincide with Table IV), the three panel means land at $\mu_{216} = 0.3214$, $\mu_{648} = 0.2623$, $\mu_{1080} = 0.2118$ with standard deviations 0.073, 0.074, and 0.090 respectively (Fig. 2). Mean δ decreases strictly monotonically with $|\mathcal{C}|$ at $\sim 5.7\sigma/\sqrt{N}$, strengthening Finding 3. At $N = 10$ this ordering inverts between $\Sigma(216 \times 3)$ and $\Sigma(360 \times 3)$ because a single outlier (`apple05`, $\delta = 0.4377$ on $\Sigma(360 \times 3)$ vs. 0.24 on the smaller groups) inflates the $\Sigma(360 \times 3)$ mean by roughly one panel standard deviation — Monte-Carlo noise visible in the small panel, resolved in the large one.

V.4. Random-diagonal ternary scan

The Tier-1 and Tier-2 panels show a striking feature: five of six structured diagonal and Howard–Vala extensions give δ values identical to four decimals across $\Sigma(72 \times 3)$

TABLE IV. Apples-to-apples Haar panel at $t = 5$: ten pre-generated single-qutrit extensions (`seed=42`) evaluated identically against each of the three base groups. The smallest δ in each row is bolded. Mean δ over the ten matrices appears in the last row; lower is better.

matrix	δ		
	$\Sigma(72 \times 3)$	$\Sigma(216 \times 3)$	$\Sigma(360 \times 3)$
<code>apple00</code>	0.2974	0.2974	0.3462
<code>apple01</code>	0.4636	0.4636	0.2029
<code>apple02</code>	0.2221	0.1727	0.2285
<code>apple03</code>	0.2493	0.2292	0.2299
<code>apple04</code>	0.3430	0.2104	0.1197
<code>apple05</code>	0.2446	0.2412	0.4377
<code>apple06</code>	0.2842	0.1701	0.1965
<code>apple07</code>	0.3169	0.2312	0.2271
<code>apple08</code>	0.3996	0.2377	0.3795
<code>apple09</code>	0.2526	0.2526	0.1823
mean	0.3073	0.2506	0.2550

and $\Sigma(216 \times 3)$. A natural question is whether this four-decimal degeneracy is specific to the cyclotomic content of the panel’s three phases $P(2\pi/9)$, $P(\pi/9)$, $P(2\pi/5)$, or whether it applies to the full maximal torus of SU(3). We answer by scanning $N = 50$ random SU(3) diagonal extensions $\text{diag}(e^{i\theta_1}, e^{i\theta_2}, e^{i\theta_3})$ (`seed = 42`, θ_1, θ_2 uniform on $[0, 2\pi)$, $\theta_3 = -\theta_1 - \theta_2 \bmod 2\pi$) on all three base groups at $t = 5$, a $3 \times 50 = 150$ -evaluation panel completed in ~ 14 minutes of wall time on a 32-core node.

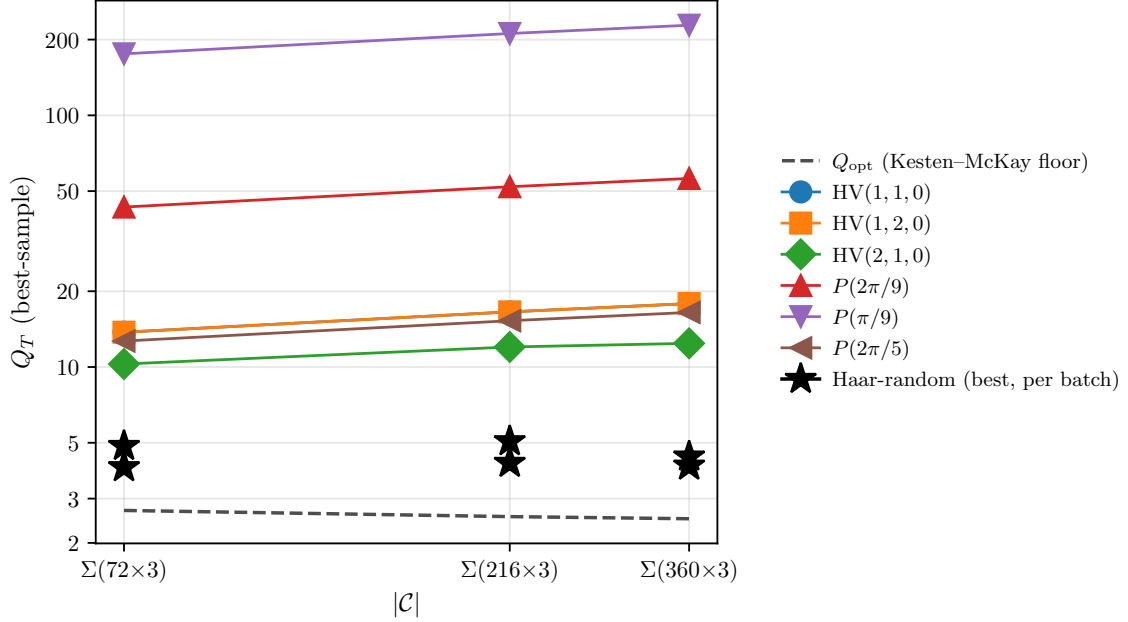


FIG. 1. Q_T across the three Σ -series base groups at $d = 3$, $t = 10$, for every extension in the panel of Table III. The dashed line is the Kesten–McKay floor Q_{opt} of Eq. (5); rational-angle diagonal and Howard–Vala extensions run nearly parallel to it at a factor ~ 3 –70 offset (Finding 2), while arbitrary-SU(3) extensions (stars) approach Q_{opt} as $|\mathcal{C}|$ grows (Finding 3). The HV(1, 1, 0) and HV(1, 2, 0) markers coincide at every $|\mathcal{C}|$ because the two gates are Clifford-equivalent at $d = 3$ (Eq. (7)).

The result is stronger than the structured-panel observation: for every one of the 50 random diagonals, the δ value on $\Sigma(72 \times 3)$ and $\Sigma(216 \times 3)$ agrees *bit-for-bit*, while $\Sigma(360 \times 3)$ is generically distinct. Per-group summary statistics (best / mean / max) across the 50 samples are

$$\Sigma(72 \times 3), \Sigma(216 \times 3) : 0.4516 / 0.6769 / 0.9921, \quad (8)$$

$$\Sigma(360 \times 3) : 0.2539 / 0.5106 / 0.9662. \quad (9)$$

Fig. 3 plots the full distribution on the SU(3) maximal-torus simplex; the Σ -series base groups have a permutation subgroup (the qutrit Clifford contains SU(3) Weyl elements), so each random-diagonal point is S_3 -symmetrized to six positions with the same δ , exposing the full torus symmetry.

The Finding-1 degeneracy is therefore *not* a property of the cyclotomic angles we chose to probe but a structural property of the nested-subgroup relation $\Sigma(72 \times 3) \subset \Sigma(216 \times 3)$ [43]: at $t = 5$, the dominant t -design irreps of SU(3) restrict to representations that both groups act on identically, so any diagonal extension couples through the same spectral channel on both. $\Sigma(360 \times 3)$ is not a supergroup of either (it contains A_5 rather than the $\Sigma(72 \times 3)$ center-extended chain [43]), hence its breakout. A $t = 10$ rerun would test whether a subdominant irrep of $\Sigma(216 \times 3)$ that is absent in $\Sigma(72 \times 3)$ becomes spectrally visible and lifts the degeneracy.

V.5. Best achievable Q_T

Combining the two panels, the best (group, extension) pair is **apple04** on $\Sigma(360 \times 3)$ at $t = 5$ with $\delta = 0.1197$ and $Q_T = 3.291$ —within 32% of $Q_{\text{opt}} = 2.495$. Note that this rnd winner, like every arbitrary-SU(d) extension, is not exactly synthesized and therefore inherits the distillation caveat of Section VII.1: the Q_T figure ranks reachability, not fault-tolerant cost. At $t = 10$ the best *fixed* extension is HV(2, 1, 0) on $\Sigma(72 \times 3)$, with $Q_T = 10.28$, a factor 3.8 above $Q_{\text{opt}} = 2.692$. The shape of the ranking is the same at both t : arbitrary-SU(3) extensions dominate every structured extension, HV(2, 1, 0) leads the structured field, and the three diagonal-phase angles $P(2\pi/9)$, $P(\pi/9)$, $P(2\pi/5)$ all give worse Q_T than even the worst Howard–Vala variant. The narrow-angle $P(\pi/9)$ in particular produces δ within 3% of unity, inflating Q_T by two orders of magnitude relative to the arbitrary-SU(3) baseline—a direct analogue of the $14 \times$ Clifford+ T penalty at $d = 2$ (Section IV).

The Tier-1 ($t = 5$) and Tier-2 ($t = 10$) panels agree to four decimals for all structured extensions except the HV(2, 1, 0) floor lift noted above; the $d = 2$ Clifford+ T $Q_T(t)$ curve of [46] reaches the same stabilization pattern by $t \gtrsim 30$, so our $t = 5/t = 10$ cross-check is the $d = 3$ analogue. The four-decimal cross-group δ -degeneracy is therefore not a finite- t artifact; it is a structural feature of rational-angle diagonal and Howard–Vala extensions on the Σ -series. Rnd batches at $t = 10$ are single-sample draws (the qco optimized kernel aggregates to one row per

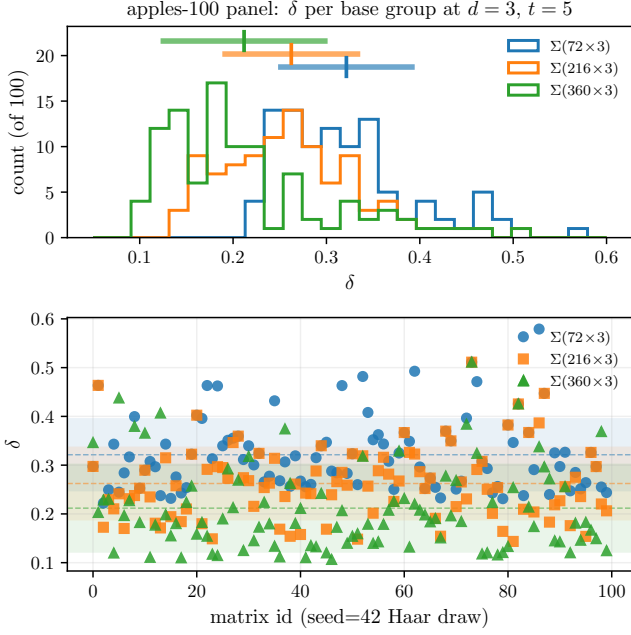


FIG. 2. Apples-to-apples Haar panel at $N = 100$ matrices on the three $d = 3$ Σ -series base groups. Matrix id $k \in \{0, \dots, 99\}$ labels `applek`, the k -th Haar-SU(3) draw from `seed = 42`; ids 0–9 coincide with the ten-matrix subset of Table IV. Top: overlaid δ histograms. Bottom: per-matrix δ for each group. The monotone reduction $\mu_{216} > \mu_{648} > \mu_{1080}$ predicted by Finding 3 is resolved at $\sim 5.7\sigma/\sqrt{N}$; the single `apple05` outlier visible as the $\Sigma(360 \times 3)$ peak near $\delta \simeq 0.44$ is the one that inverted the ten-matrix ordering in Table IV.

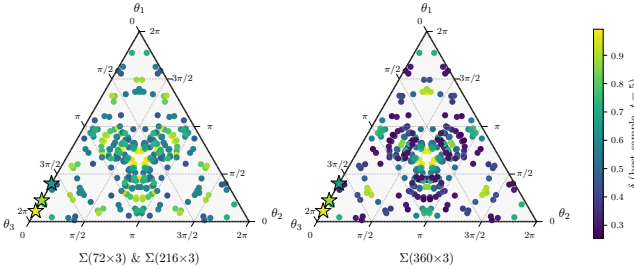


FIG. 3. δ on the SU(3) maximal torus at $t = 5$. Each panel shows the 50 random-diagonal extensions of Section V.4, S_3 -symmetrized to six copies each (300 points per panel). The left panel, labeled $\Sigma(72 \times 3)$ & $\Sigma(216 \times 3)$, reports the δ values that are bit-for-bit identical on the two smaller Σ -series groups; the right panel reports $\Sigma(360 \times 3)$. Red stars mark the three structured rational-angle extensions $P(2\pi/9)$, $P(\pi/9)$, $P(2\pi/5)$ from Table III, colored by their own δ on each group; they sit on the $\theta_1 = 0$ edge because each has the form $\text{diag}(1, e^{i\theta}, e^{-i\theta})$.

subprocess regardless of the `sample_size` argument); statistically significant Haar statistics come from the apples-100 panel in Section V.3 and from the 10^4 -sample BI pilot at $d = 2$ (Section IV).

VI. THE DSA PRIMITIVE-GATE TARGET FAMILY

The $d = 3$ survey of Section V ranks extensions by the Haar-uniform figure of merit Q_T . Many applications do not need a uniform net. Discrete-subgroup approximations of lattice gauge simulations [29, 48, 55] require instead a small structured family of unitaries: a few continuous-angle R_Z rotations with group-theoretic coefficients, plus a finite set of cyclotomic phases from the discrete Fourier transform U_F . On such a workload the relevant figure of merit is the mean synthesis cost over that finite family, not a t -design spectral gap. The two figures of merit disagree in general. We note that $\Sigma(36 \times 3)$ itself appears here as a *target family* of SU(2) rotations derived from its trace spectrum under a qubit encoding [30]; it is not used as a base group, its reducible adjoint at $d = 3$ having excluded it from the QCO survey (Section V).

A concrete target family. We work out the case of the $\Sigma(36 \times 3)$ encoding as a representative example; different subgroups in the series produce different coefficient sets. The trace gate $U_{\text{Tr}}(\theta) = \text{diag}_{g \in \mathcal{C}}(e^{i\theta \text{Re Tr}(g)})$ appears in the magnetic plaquette of any DSA lattice gauge simulation [29]. For $\Sigma(36 \times 3)$, Gustafson *et al.* [30] reduce it to a three-qubit trace register: the 14 conjugacy classes collapse to 7 distinct $\text{Re Tr}(g) \in \{3, \pm 1, \pm \frac{1}{2}, -\frac{3}{2}, 0\}$, and a Boolean map $(p, q, r, s, t) \mapsto (v_0, v_1, v_2)$ separates them. On the (v_0, v_1, v_2) register the trace Hamiltonian factorizes as

$$H_{\text{Tr}} = \frac{1}{16} [3(III + IIZ + IZI) + 5 IZZ + 1 ZII + 9(ZIZ + ZZI) + 15 ZZZ] \quad (10)$$

so $U_{\text{Tr}}(\theta)$ decomposes (up to Clifford conjugation) into R_Z rotations at angles proportional to $\alpha_c \theta$ for $\alpha_c \in \{1, 3, 5, 9, 15\}$ — the five distinct non-zero coefficients in the decomposition. These coefficients are specific to $\Sigma(36 \times 3)$; other subgroups in the series produce different trace spectra and different α_c . The discrete half comes from U_F : up to global phase, the Howard–Vala qutrit gate $\text{HV}(1, 2, 0)$ [38] injects 9th-root phases, requiring $R_Z(2\pi k/9)$ for $k = 1, \dots, 8$. The synthesis target for the $\Sigma(36 \times 3)$ qubit pipeline is therefore

$$\mathcal{T}_{36} = \{R_Z(2\pi k/9)\}_{k=1}^8 \cup \{R_Z(\alpha_c \theta)\}_{\alpha_c \in \{1, 3, 5, 9, 15\}, \theta \in (0, \pi)}. \quad (11)$$

Mean-depth figure of merit. Given a qubit base group $\mathcal{C}$ and single-gate extension g , SWIFTBOT enumerates the word tree of $\mathcal{C} \cup \{g, g^{-1}\}$ by breadth-first search with projective-phase deduplication, capped at 2×10^5 unique words. For each $U \in \mathcal{T}_{36}$ we record the shortest BFS word whose projective distance $d(U, W) = \sqrt{2 - |\text{Tr}(U^\dagger W)|}$ falls below $\varepsilon = 10^{-2}$, or the depth of closest approach if no such word exists. The $\varepsilon = 10^{-2}$ floor matches the per-link accuracy typical of near-term Trotter budgets [29]; the BFS-exhaust cap prevents runaway search at this tolerance, and tightening ε can only increase the $6.8\times$ advantage (see §Caveats below). Discrete targets may be hit *exactly* at finite depth, in which case the corresponding

TABLE V. DSA $\Sigma(36 \times 3)$ target-family coverage at BFS depth 15, $\varepsilon = 10^{-2}$, 10 continuous θ -samples per α_c . “hits” counts the 58 targets (the 8 discrete 9th-root phases in Eq. (11) plus 50 continuous $R_Z(\alpha_c \theta)$ samples) reached below ε . “mean/max” are projective distances across all 58; lower is better. $\langle T \rangle$ is the mean T -count along hit words. “method” is **rs** for Ross–Selinger [10] exact synthesis and **bfs** for the SWIFTBOT BFS estimate; all rows are **bfs** except the canonical $P(\pi/4)$, no RS-style normal form having yet been established for the rational-angle cyclotomic rings $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$ or $\mathbb{Z}[i, \sqrt{2}]$.

Base + ext.	hits (/58)	mean dist.	max dist.	$\langle T \rangle$	method
Clifford+ $P(\pi/18)$	34	0.0104	0.0420	2.79	bfs
hurwitz + $P(\pi/18)$	35	0.0112	0.0420	5.37	bfs
Clifford+ $P(\pi/9)$	27	0.0120	0.0401	2.04	bfs
Clifford+ $P(2\pi/9)$	27	0.0122	0.0401	2.11	bfs
Clifford+ $P(\pi/8)$	7	0.0221	0.0387	0.29	bfs
Clifford+ T_{24}	6	0.0228	0.0465	0.83	bfs
Clifford+ T_{12}	8	0.0233	0.0474	1.88	bfs
hurwitz + T_{12}	0	0.0313	0.0573	—	bfs
Clifford+ $P(\pi/4)$	5	0.0833	0.1554	0.00	rs

distance is zero and the T -count is read directly from the word. Continuous targets are sampled at ten uniform $\theta \in (0, \pi)$ per α_c , for fifty continuous evaluations per extension. We report mean distance across Eq. (11) as the single headline number; ties are broken on the discrete-exact count.

Result. Table V ranks the candidates at BFS depth 15, $\varepsilon = 10^{-2}$, $N_\theta = 10$ per continuous coefficient. The Clifford+ $P(2\pi/9)$ extension attains mean projective distance 0.0122 on $\mathcal{T}_{36}$ with 27 of 58 targets hit below ε , while the canonical Clifford+ T extension attains mean distance 0.0833 with only 5 hits—a factor $6.8\times$ separation in mean distance. The equivalence class of $P(2\pi k/18)$ with $\gcd(k, 18) = 1$ (which includes $P(\pi/9)$ and $P(\pi/18)$) all saturate the discrete 9th-root targets, a direct consequence of $\langle \text{Clifford}, P(2\pi/9) \rangle$ closing at a finite group of $\sim 3 \times 10^4$ elements; the algebraic ring $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$ supplies every 9th-root phase via a short word. Within this equivalence class, Clifford+ $P(\pi/18)$ marginally edges Clifford+ $P(2\pi/9)$ at depth 15 (mean 0.0104 vs. 0.0122; 34 vs. 27 hits), reflecting the strictly larger 18th-root coverage: $P(\pi/18)$ generates ω_{36} , which contains both the 9th- and 18th-root phases of U_F . Crucially, the super-golden extensions T_{24} and T_{12} —which optimize Q_T for Clifford and for the 12-element projective A_4 base **hurwitz** respectively [47]—perform *worse* on $\mathcal{T}_{36}$ than the rational-angle rows: their algebra $\mathbb{Z}[i, \sqrt{2}]$ does not contain ω_9 , so they cannot reach the 9th-root targets exactly at any depth; on the $|\mathcal{C}| = 12$ **hurwitz** base the T_{12} extension hits zero targets at depth 15. The **hurwitz**+ $P(\pi/18)$ row is competitive (35 hits, mean 0.0112) and does so with a strictly *smaller* base group: in the target-family regime the overhead advantage tracks the cyclotomic content of the extension, not the size of the base.

Interpretation. Clifford+ $P(2\pi/9)$ is worse on Q_T than Clifford+ T yet $6.8\times$ better on mean depth over $\mathcal{T}_{36}$. The

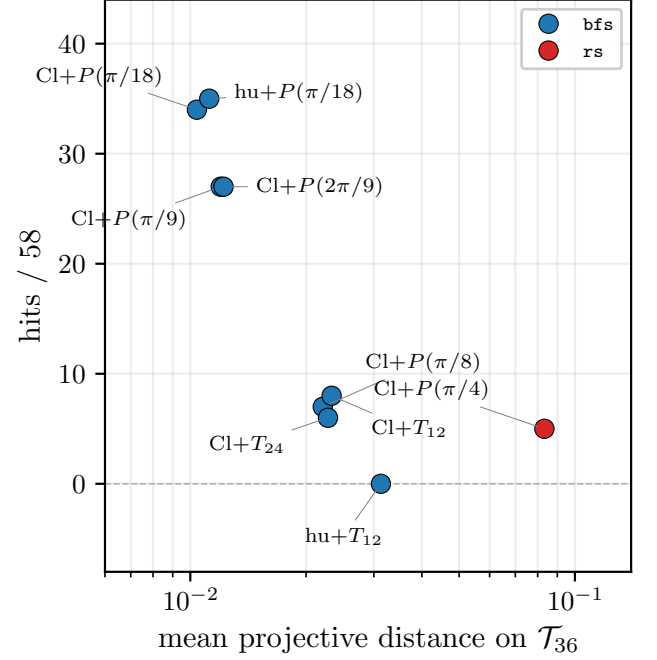


FIG. 4. DSA $\Sigma(36 \times 3)$ target-family coverage from Table V: hits/58 versus mean projective distance on $\mathcal{T}_{36}$ at BFS depth 15, $\varepsilon = 10^{-2}$. Canonical Clifford+ T (red, **rs_exact** certified) sits bottom-right with the largest mean distance and only 5 hits. Rational-angle $P(k\pi/18)$ extensions (blue, **bfs_estimate**) cluster top-left with many hits at an order of magnitude smaller distance; super-golden T_{12}/T_{24} rows fall in between despite saturating Q_{opt} (Section IV). The visual separation matches the $6.8\times$ headline.

reason is structural: Q_T is the coefficient of a Haar-uniform covering rate, while $\mathcal{T}_{36}$ is a measure-zero algebraic slice of $\text{SU}(2)$ whose ω_9 cyclotomic content matches $\langle \text{Clifford}, P(2\pi/9) \rangle$. The super-golden extensions saturate Q_{opt} but equidistribute in the wrong direction for this target. A lattice-gauge simulator compiling Eq. (11) should therefore rank extensions by target-family mean depth, not by Q_T , and the two rankings can disagree by nearly an order of magnitude.

Caveats. The Clifford+ T row’s T -count is certified by Ross–Selinger exact synthesis [10, 11]; the companion Clifford-cyclotomic framework of Forest *et al.* [59] extends this to $\mathbb{Z}[\zeta_{2^n}]$ but not to the rational-prime-denominator rings relevant here. No analogous normal form exists for $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$, so the rational-angle rows carry depth-15 BFS upper bounds. The $6.8\times$ advantage is therefore a *lower bound*: a certified synthesis theorem over $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$ would only shorten the BFS-side words, while the canonical Ross–Selinger side is already tight. Anderson–Jochym–O’Connor [18] excludes any qubit *stabilizer* code with transversal $|k\pi/9\rangle$. Two routes sidestep this. Programmable MSD [60–62] is indirect synthesis: it prepares $|\psi(\theta)\rangle$ from already-distilled $|T\rangle$ on a Clifford+ T stabilizer substrate, so the AJOC no-go never applied to

it. Non-stabilizer code-switching to a Kubischta–Teixeira PI code [5, 63] is a genuine bypass but requires a PI code of sufficient distance hosting $|\omega_9\rangle$, which has not been constructed. The $6.8\times$ advantage is cleanest when $P(2\pi/9)$ is a native primitive—e.g., in a qutrit register or on a hypothetical nonadditive code that distills $|\omega_9\rangle$ directly.

VII. DISCUSSION

Two findings drive the discussion below. First, at $d = 3$ and $t = 5$, arbitrary-SU(3) extensions on the largest irreducible-adjoint Σ -series base reach $Q_T/Q_{\text{opt}} \approx 1.32$ —essentially the same “golden-gate direction” observed at $d = 2$, where the arbitrary-SU(2) baseline sits 4% above Q_{opt} on the 12-element *hurwitz* base (Section IV). No structured extension in our panel (Howard–Vala or rational diagonal) gets closer than a factor ~ 2.9 of Q_{opt} . Second, and in apparent tension with the first, the non-canonical Clifford+ $P(2\pi/9)$ extension beats the canonical Clifford+ T by $6.8\times$ on the DSA primitive-gate target family (Section VI) despite being $\sim 8\times$ worse on Q_T . The two findings together are the paper’s central design lesson: the “best general-purpose extension” and the “best target-specific extension” optimize different objectives, and systematic qudit fault-tolerance design must report both.

VII.1. Fault-tolerance status of the winning extensions

A Q_T ranking is necessary but not sufficient for fault-tolerant implementation. Table VI audits every extension evaluated against the curated SWIFTBOT catalogs of transversal codes, distillation protocols, and certified synthesis rings. Only canonical Clifford+ T at $d = 2$ admits an end-to-end certified stack and serves as our validation anchor. Three rows inherit a transversal-code half only: *hurwitz*-based super-golden extensions via Parzanchevski–Sarnak [54], BI+*rnd* via the Kubischta 2I code [6], and the $d = 3$ Σ +HV(z, γ, ε) rows via the qutrit triorthogonal $[[9m - k, k, 2]]_3$ family [23]. All other rows lack either a distillation circuit or a certified synthesis ring.

Three regimes deserve comment. First, the cyclotomic ring $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$ —hosting the $d = 2$ target-family winners $P(2\pi/9)$, $P(\pi/9)$, $P(\pi/18)$ —has no Ross–Selinger-style normal form, and the two non-stabilizer handling routes discussed in Section VI Caveats (indirect programmable MSD on a Clifford+ T substrate; PI code-switching pending a suitable distance-preserving PI code) remain the best-known options at $d = 2$. At $d = 3$ the same rotations are qutrit-native and Campbell–Anwar–Browne [23] distills $|\omega_9\rangle$ directly.

Second, the super-golden gates of Parzanchevski–Sarnak [54] live in $\mathbb{Z}[i, \sqrt{2}]$ and are not Clifford-orbit rotations of stabilizer states, so triorthogonal distillation does not apply. Whether a golden-gate-compatible dis-

tillation circuit can be built—possibly on the Kubischta 2I code, whose transversal group is precisely the super-golden base—is open.

Third, arbitrary-SU(d) extensions are by definition not exactly synthesized, and so cannot be distilled in the usual sense [2, 3]. This applies uniformly to every *rnd* row; the best current claim is transversal-group support at the base-code level.

For the Σ -series at $d = 3$, data and distillation codes are structurally separable. The Howard–Vala magic state is distilled by the qutrit triorthogonal family [23, 24], but no explicit data code hosts a Σ -series group transversally. Denys–Leverrier covariant codes at $d = 2$ [7, 8] support $2T$ and $2O$ group actions natively through continuous-group covariance, distinct from Eastin–Knill transversality on a stabilizer code; extending either to a qutrit $\Sigma(m \times 3)$ base—an open case inside the twisted unitary t -group program [63]—would complete the stack. (Clifford+ R synthesis [45] separately closes the compilation-certification gap for these $d = 3$ gate sets; we return to this below.)

Higher-dimensional qudit MSD is advancing rapidly. Krishna–Tillich [64] opened the low-overhead program; Wills, Hsieh, and Yamasaki [25] achieved the first constant-overhead MSD construction via asymptotically good quantum codes; Golowich–Guruswami [65] and Nguyen [66] gave asymptotically good qubit codes with transversal non-Clifford and CCZ gates. Most directly applicable to our framework, Cervia *et al.* [26] construct triorthogonal codes on $\mathbb{F}_{2^{2m}}$ at qudit dimensions as small as $q = 64$ that beat the Tsfasman–Vlăduț–Zink bound and give constant-space-overhead $|CCZ\rangle$ distillation, together with an explicit qudit-to-qubit conversion circuit that makes the construction a drop-in upgrade for any Clifford+CCZ evaluation. None of these high- d codes have cataloged transversal pairings with our base-group registry, but the infrastructure now exists to pair them.

A complementary recent development is *magic-state cultivation* [4]: rather than distilling high-fidelity $|T\rangle$ states from many noisy copies, one grows a protected $|T\rangle$ inside a surface-code patch using roughly the same gate budget as a logical CNOT, reducing qubit-round cost by an order of magnitude at realistic noise [27], with an experimental demonstration on a superconducting processor [28]. Cultivation is currently qubit-native and surface-code-specific; the open question for the qudit program is whether a native- d analogue exists on codes with transversal non-Clifford structure (Kubischta 2I, Denys–Leverrier $2T/2O$, or the twisted t -group codes of [63]). Two arguments have historically favored $d = 2$: the maturity of Ross–Selinger Clifford+ T synthesis and the qubit-specificity of magic-state cultivation. The first is closing at $d = 3$ with Clifford+ R synthesis [45]; a qudit cultivation analogue would close the second.

On the synthesis side of the qudit story, single-qutrit exact-synthesis methods over Clifford+ T and related Clifford+cyclotomic rings are by now well developed [45, 67–69], so the qutrit compilation story is closing to parity with the qubit one.

TABLE VI. Magic-state distillation (MSD) status for every extension evaluated in this paper. “Algebra” is the cyclotomic or quaternionic ring hosting the non-Clifford factor of the extension. “Status/closest cousin” combines synthesis and distillation coverage: COMPLETE means transversal code, published MSD protocol, and certified normal-form synthesis are all available; PARTIAL means a transversal code is available but no matching MSD circuit (or the code–extension pairing is not yet established); NONE means no cataloged MSD, either because no protocol exists for the algebra or because the extension is not exactly synthesizable. Rows are grouped by qudit dimension and by the panel section in which the extension appears.

Extension	Algebra	Status / closest cousin	Reference
<i>d = 2 canonical (Section IV)</i>			
Clifford+ $P(\pi/4)$ (canonical $Z[\omega_8, 1/\sqrt{2}]$)		COMPLETE: BK 15-to-1; Bravyi–Haah triorthogonal	[2, 3]
Clifford+ $P(\pi/8)$	$Z[\omega_{16}, 1/2]$	NONE: no published protocol for ω_{16} ; cousin BK 15-to-1 [2] does not distill this state	
<i>d = 2 DSA target family (Section VI)</i>			
Clifford+ $P(2\pi/9)$, $P(\pi/9)$, $Z[\omega_{18}, 1/2]$ $P(\pi/18)$		PARTIAL (INDIRECT): no direct stabilizer-code distillation (Anderson–Jochym–O’Connor [18]); programmable-MSD distills $ \psi(\theta)\rangle$ on a Clifford+ T substrate; PI code-switching [5] is a non-stabilizer alternative	[5, 60–62]
Clifford+ $P(2\pi/5)$	$Z[\omega_{10}]$	PARTIAL (INDIRECT): stabilizer-code distillation excluded [18]; programmable-MSD and PI code-switching both apply to arbitrary θ	[5, 60–62]
Clifford+ T_{24} , hurwitz+ T_{12} (super-golden)	$Z[i, \sqrt{2}]$	PARTIAL: Parzanchevski–Sarnak supplies transversal base; magic state is not a Clifford-orbit stabilizer rotation, so triorthogonal MSD does not apply	[54]
<i>Arbitrary-SU(d) rows, any d</i>			
Clifford+rnd, hurwitz+rnd ($d = 2$)	generic SU(2) (no ring)	NONE: not exactly synthesizable, so MSD is not defined	—
BI+rnd ($d = 2$)	generic SU(2) (no ring)	PARTIAL: Kubischta ((7, 2, 3)) supplies transversal $2I$ base; no MSD for a Haar magic state	[6]
Σ -series+rnd ($d = 3$)	generic SU(3) (no ring)	NONE: not exactly synthesizable; no cataloged qutrit transversal classification for Σ -series	—
<i>d = 3 structured (Section V)</i>			
$\Sigma + \text{HV}(1, 1, 0)$, $Z[\omega_9]$		PARTIAL: qutrit triorthogonal $[[9m-k, k, 2]]_3$ distills Howard–Vala magic state; Σ -series transversal pairing unestablished	[23]
$\Sigma + \text{HV}(1, 2, 0)$		PARTIAL: cousin Campbell–Anwar–Browne qutrit triorthogonal; no protocol certified for this orbit	[23]
$\Sigma + \text{HV}(2, 1, 0)$	$Z[\omega_9]$	PARTIAL: qutrit-native ω_9 phase; Campbell–Anwar–Browne applies modulo transversal pairing	[23]
$\Sigma + P(2\pi/9)$ ($d = 3$)	$Z[\omega_9]$	NONE: ω_{18} outside Campbell–Anwar–Browne; no cousin at $d = 3$	—
$\Sigma + P(\pi/9)$ ($d = 3$)	$Z[\omega_{18}]$	NONE: ω_{10} not in any qutrit stabilizer setting; no cousin	—
$\Sigma + P(2\pi/5)$ ($d = 3$)	$Z[\omega_{10}]$		

VII.2. Limitations

Three limitations frame the numerical claims. First, the $d = 3$ Haar statistics come from either single-sample $t = 10$ draws (Tier-2 panel) or the 100-matrix apples-to-apples panel at $t = 5$ (Section V.3); a joint $t = 10$, ~ 100 -sample Haar run is held up by a qco-kernel limitation that aggregates each subprocess to a single output row regardless of sample size. Second, no $d = 4$ survey is attempted: largest-irrep dimensions scale as t^6 at $d = 4$ and squared matrix-storage as t^{12} , so only the smallest SU(4) subgroups ($|C| \lesssim 300$) fit in a single-node 125 GB budget at $t = 10$. A pilot on those is tractable overnight; the large- $|C|$ SU(4) subgroups (e.g. S7F) would need a cluster slot. Third, the target-family T -counts in Table V are depth-15 BFS upper bounds except for the Ross–Selinger-certified Clifford+ $P(\pi/4)$ row. The $6.8\times$ advantage is a *lower bound*: a certified normal form over $Z[\omega_{18}, \frac{1}{2}]$ [59] would only tighten the BFS side, not the

canonical side, which is already exact.

VII.3. Open questions

Two quantitative questions are directly posed by our $t = 5$ data and should be sharp enough to settle at $t \geq 10$: (a) does the four-decimal δ -degeneracy across $\Sigma(216, 648, 1080 \times 3)$ observed for structured diagonal and Howard–Vala extensions in Table III persist, break, or re-rank at larger t ? and (b) does the $\Sigma(360 \times 3)$ apples-panel mean cross strictly below the $\Sigma(216 \times 3)$ mean once the ~ 100 -matrix apples panel removes the 10-matrix outlier sensitivity? Question (b) has been *settled in the affirmative* at $t = 5$ in Section V.3 ($5.7\sigma/\sqrt{N}$ separation at $N = 100$); the $t \geq 10$ rerun will establish whether the same ordering holds at converged design order. Three qualitative questions are left open: (c) a super-golden construction at $d = 3$ saturating $Q_{\text{opt}} \approx 2.69$ on $\Sigma(72 \times 3)$,

2.54 on $\Sigma(216 \times 3)$, or 2.50 on $\Sigma(360 \times 3)$; (d) an explicit data code C_{data} that hosts one of our Σ -series base groups transversally, analogous to the Denys–Leverrier constructions for $2T$ and $2O$ at $d = 2$ [7, 8] or to the Kubischta–Teixeira twisted unitary t -group framework [63]; and (e) a Ross–Selinger-style normal form for the cyclotomic ring $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$ that would certify the Table V T -counts.

VII.4. Software release

SWIFTBOT is released with this paper [49] under the MIT license. The release contains: the five-stage pipeline (Section III); the pre-seeded group registry for $d \in \{2, 3, 4\}$; the SWIFTBOT optimizations of the QCO kernel of [46], including the conjugation cache and on-disk $\pi_\lambda(\mathcal{C})$ cache described in Section III; the Stage-4 QEC-code catalog and Stage-5 distillation-protocol catalog with their cross-references to transversal groups; the BFS target-family coverage module used in Section VI; the `q-panel` CLI entry point and the two panel specifications `d3_survey` and `d3_rnd_apples` that reproduce Tables III and IV bit-identically; the SQLite cache containing all 54 per-sample rows quoted in Section V; and the L^AT_EX source of this paper. A minimal README walks new users through reproducing Tables II to IV on a laptop in under an hour. Pull requests adding new base groups, extensions, codes, or distillation protocols are welcome; the catalog cross-validation tests ensure that every registered transversal-group string resolves to a registered base group.

VIII. CONCLUSIONS

SWIFTBOT—an agentic workflow that ranks $(\mathcal{C}, g)$ pairs by Q_T and cross-references each against a curated fault-tolerance catalog—produces the first systematic $d = 3$ Q_T ranking in our registry and identifies $\Sigma(360 \times 3)$ with an arbitrary-SU(3) extension as the best

general-purpose qudit base ($Q_T/Q_{\text{opt}} \approx 1.32$ in the apples-100 panel at $t = 5$, and consistent single-sample draws at $t = 10$). A systematic 10^4 -sample arbitrary-SU(2) pilot on the binary icosahedral base at $d = 2$ gives $Q_T/Q_{\text{opt}} \approx 1.27$, the natural super-golden candidate. Applied to a lattice-gauge simulation target family, the non-canonical Clifford+ $P(2\pi/9)$ beats canonical Clifford+ T by $6.8\times$ on mean depth despite being worse on Q_T , establishing that the two rankings locate different gates. Only canonical Clifford+ T admits an end-to-end certified fault-tolerant stack; every other winner is covered partially, with sharp open algebraic-compilation or code-pairing problems standing between it and full fault-tolerance. Three concrete next steps carry most of the remaining design risk. (i) A $d = 4$ pilot on the Fairbairn subgroups [40] would test whether the $Q_T/Q_{\text{opt}} \rightarrow 1$ trend continues past $d = 3$ or the $d = 3$ “golden direction” is an accident. (ii) An explicit Σ -series $\leftrightarrow$ data-code transversal pairing [5, 8, 63] would upgrade the $d = 3$ winners from PARTIAL to COMPLETE in Table VI. (iii) A Ross–Selinger normal form for $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$ would replace the BFS upper bound in Table V and likely sharpen the $6.8\times$ headline. For a hardware vendor choosing between qubit and qudit architectures *today*, the practical takeaway is narrower: if the workload is lattice-gauge-simulation-shaped, a rational-phase qubit Clifford extension already buys nearly an order of magnitude over the canonical Clifford+ T stack with no new hardware.

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- [1] B. Eastin and E. Knill, Restrictions on Transversal Encoded Quantum Gate Sets, *Phys. Rev. Lett.* **102**, 110502 (2009), [arXiv:0811.4262 \[quant-ph\]](#).
 - [2] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, *Phys. Rev. A* **71**, 022316 (2005), [arXiv:quant-ph/0403025](#).
 - [3] S. Bravyi and J. Haah, Magic-state distillation with low overhead, *Phys. Rev. A* **86**, 052329 (2012), [arXiv:1209.2426](#).
 - [4] C. Gidney, N. Shutty, and C. Jones, Magic state cultivation: growing T states as cheap as CNOT gates (2024), [arXiv:2409.17595 \[quant-ph\]](#).
 - [5] Y. Ouyang, Y. Jing, and G. K. Brennen, Measurement-free code-switching for low overhead quantum computation using permutation invariant codes, *arXiv preprint* (2024), [arXiv:2411.13142 \[quant-ph\]](#).
 - [6] E. Kubischta and I. Teixeira, Family of Quantum Codes with Exotic Transversal Gates, *Phys. Rev. Lett.* **131**, 240601 (2023), [arXiv:2305.07023 \[quant-ph\]](#).
 - [7] A. Denys and A. Leverrier, The $2T$ -qutrit, a two-mode bosonic qutrit, *Quantum* **7**, 1032 (2023), [arXiv:2210.16188 \[quant-ph\]](#).
 - [8] A. Denys and A. Leverrier, Quantum error-correcting codes with a covariant encoding, *Phys. Rev. Lett.* **133**, 240603 (2024), [arXiv:2306.11621 \[quant-ph\]](#).

- [9] A. Y. Kitaev, Quantum computations: algorithms and error correction, *Russ. Math. Surv.* **52**, 1191 (1997).
- [10] N. J. Ross and P. Selinger, Optimal ancilla-free Clifford+T approximation of z -rotations, *Quant. Inf. Comp.* **16**, 901 (2016), [arXiv:1403.2975 \[quant-ph\]](#).
- [11] V. Kliuchnikov, D. Maslov, and M. Mosca, Fast and efficient exact synthesis of single qubit unitaries generated by Clifford and T gates, *Quant. Inf. Comp.* **13**, 607 (2013), [arXiv:1206.5236 \[quant-ph\]](#).
- [12] M. Ringbauer, M. Meth, L. Postler, R. Stricker, R. Blatt, P. Schindler, and T. Monz, A universal qudit quantum processor with trapped ions, *Nature Phys.* **18**, 1053 (2022), [arXiv:2109.06903 \[quant-ph\]](#).
- [13] T. Kim, T. Roy, X. You, A. C. Y. Li, H. Lamm, O. Pronitchev, *et al.*, Ultracoherent superconducting cavity-based multiqudit platform with error-resilient control (2025), [arXiv:2506.03286 \[quant-ph\]](#).
- [14] L. Pecorari, S. Jandura, G. K. Brennen, and G. Pupillo, High-rate quantum LDPC codes for long-range-connected neutral atom registers, *Nature Commun.* **16**, 1111 (2025), [arXiv:2404.13010 \[quant-ph\]](#).
- [15] Y. Wang, Z. Hu, B. C. Sanders, and S. Kais, Qudits and High-Dimensional Quantum Computing, *Front. Phys.* **8**, 589504 (2020).
- [16] E. O. Kiktenko, A. S. Nikolaeva, and A. K. Fedorov, Colloquium: Quantum Computing with Qudits, *Rev. Mod. Phys.* **97**, 021003 (2025), [arXiv:2212.12934 \[quant-ph\]](#).
- [17] D. Gottesman, Fault-tolerant quantum computation with higher-dimensional systems, *Chaos, Solitons & Fractals* **10**, 1749 (1999), [arXiv:quant-ph/9802007](#).
- [18] J. T. Anderson and T. Jochym-O'Connor, Classification of transversal gates in qubit stabilizer codes, *Quant. Inf. Comp.* **16**, 0771 (2016), [arXiv:1409.8320 \[quant-ph\]](#).
- [19] D. M. K rk og lu, H. Lamm, O. Ogunkoya, and L. Pierattelli, The $2T$ -quocit: a two-mode bosonic qudit for high energy physics, *arXiv preprint* (2025), [arXiv:2509.21941 \[quant-ph\]](#).
- [20] S. Bravyi, A. W. Cross, J. M. Gambetta, D. Maslov, P. Rall, and T. J. Yoder, High-threshold and low-overhead fault-tolerant quantum memory, *Nature* **627**, 778 (2024), [arXiv:2308.07915 \[quant-ph\]](#).
- [21] D. J. Spencer, A. Tanggara, T. Haug, D. Khu, and K. Bharti, Qudit low-density parity-check codes, *Quantum* **10**, 2023 (2026), [arXiv:2510.06495 \[quant-ph\]](#).
- [22] B. Vasi , V. Savin, M. Pacenti, S. Borah, and N. Raveendran, Quantum Low-Density Parity-Check Codes (2025), [arXiv:2510.14090 \[quant-ph\]](#).
- [23] E. T. Campbell, H. Anwar, and D. E. Browne, Magic-state distillation in all prime dimensions using quantum Reed-Muller codes, *Phys. Rev. X* **2**, 041021 (2012), [arXiv:1205.3104](#).
- [24] S. Prakash and T. Saha, Low-overhead qutrit magic state distillation, *Quantum* **9**, 1768 (2025), [arXiv:2403.06228 \[quant-ph\]](#).
- [25] A. Wills, M.-H. Hsieh, and H. Yamasaki, Constant-overhead magic state distillation, *Nature Physics* **10.1038/s41567-025-03026-0** (2025).
- [26] M. J. Cervia, H. Lamm, D. Liu, E. M. Muraire, and S. Zhu, Magic State Distillation using Asymptotically Good Codes on Qudits (2025), [arXiv:2512.21874 \[quant-ph\]](#).
- [27] Y. Vakin, S. Jacoby, A. Grimsmo, and A. Retzker, Efficient Magic State Cultivation on the Surface Code (2025), [arXiv:2502.01743 \[quant-ph\]](#).
- [28] E. Rosenfeld *et al.*, Magic state cultivation on a superconducting quantum processor (2025), [arXiv:2512.13908 \[quant-ph\]](#).
- [29] H. Lamm, S. Lawrence, and Y. Yamauchi (NuQS), General methods for digital quantum simulation of gauge theories, *Phys. Rev. D* **100**, 034518 (2019), [arXiv:1903.08807 \[hep-lat\]](#).
- [30] E. J. Gustafson, Y. Ji, H. Lamm, E. M. Muraire, S. Osorio Perez, and S. Zhu, Primitive Quantum Gates for an $SU(3)$ Discrete Subgroup: $\Sigma(36 \times 3)$, *Phys. Rev. D* **110**, 034515 (2024), [arXiv:2405.05973 \[hep-lat\]](#).
- [31] M. Meth, J. F. Haase, J. Zhang, C. Edmunds, L. Postler, A. Steiner, A. J. Jena, L. Dellantonio, R. Blatt, P. Zoller, T. Monz, P. Schindler, C. Muschik, and M. Ringbauer, Simulating 2D lattice gauge theories on a qudit quantum computer, *Nature Phys.* **21**, 570 (2025), [arXiv:2310.12110 \[quant-ph\]](#).
- [32] P. P. Popov, M. Meth, M. Lewenstein, P. Hauke, M. Ringbauer, E. Zohar, and V. Kasper, Variational quantum simulation of $U(1)$ lattice gauge theories with qudit systems (2023), [arXiv:2307.15173 \[quant-ph\]](#).
- [33] L. Spagnoli, A. Roggero, and N. Wiebe, Qudit stabiliser codes for $\mathbb{Z}_N$ lattice gauge theories with matter (2026), [arXiv:2602.20661 \[quant-ph\]](#).
- [34] Z. P. Bradshaw, Approximate Error Correction for Quantum Simulations of $SU(2)$ Lattice Gauge Theories (2026), [arXiv:2603.26819 \[quant-ph\]](#).
- [35] J. N. Eberhardt and V. Steffan, Logical Operators and Fold-Transversal Gates of Bivariate Bicycle Codes (2024), [arXiv:2407.03973 \[quant-ph\]](#).
- [36] H. Sayginel, S. Koutsoumpas, M. Webster, A. Rajput, and D. E. Browne, Fault-Tolerant Logical Clifford Gates from Code Automorphisms (2024), [arXiv:2409.18175 \[quant-ph\]](#).
- [37] A. J. Malcolm, A. N. Glaudell, P. Fuentes, D. Chandra, A. Schotte, C. DeLisle, R. Haenel, A. Ebrahimi, J. Roffe, A. O. Quintavalle, S. J. Beale, N. R. Lee-Hone, and S. Simmons, Computing Efficiently in QLDPC Codes (2025), [arXiv:2502.07150 \[quant-ph\]](#).
- [38] M. Howard and J. Vala, Qudit versions of the qubit $\pi/8$ gate, *Phys. Rev. A* **86**, 022316 (2012), [arXiv:1206.1598 \[quant-ph\]](#).
- [39] H. F. Blichfeldt, *Finite Collineation Groups* (University of Chicago Press, 1917).
- [40] W. M. Fairbairn, T. Fulton, and W. H. Klink, Finite and Disconnected Subgroups of $SU(3)$ and their Application to the Elementary-Particle Spectrum, *J. Math. Phys.* **5**, 1038 (1964).
- [41] H. Flyvbjerg, Character table for the 1080 element point group like subgroup of $SU(3)$, *J. Math. Phys.* **26**, 2985 (1985).
- [42] A. Hanany and Y.-H. He, A Monograph on the Classification of the Discrete Subgroups of $SU(4)$, *JHEP* **02**, 027, [arXiv:hep-th/9905212](#).
- [43] W. Grimus and P. O. Ludl, Principal series of finite subgroups of $SU(3)$, *J. Phys. A* **43**, 445209 (2010), [arXiv:1006.0098 \[math-ph\]](#).
- [44] A. Bocharov, M. Roetteler, and K. M. Svore, Factoring with qutrits: Shor's algorithm on ternary and metaplectic quantum architectures, *Phys. Rev. A* **96**, 012306 (2017), [arXiv:1605.02756 \[quant-ph\]](#).
- [45] E. J. Gustafson, H. Lamm, D. Liu, E. M. Muraire, and S. Zhu, Synthesis of Single Qutrit Circuits from Clifford+ R (2025), [arXiv:2503.20203 \[quant-ph\]](#).

- [46] O. Słowik, P. Dulián, and A. Sawicki, Quantum Circuit Overhead: the cost of approximating unitaries with finite gate sets, arXiv preprint (2025), [arXiv:2505.00683](https://arxiv.org/abs/2505.00683) [quant-ph].
- [47] P. Sarnak, [Letter to Aaronson and Pollington on the Solovay–Kitaev theorem and golden gates](#) (2015).
- [48] E. J. Gustafson, H. Lamm, F. Lovelace, and D. Musk, Primitive Quantum Gates for an $SU(2)$ Discrete Subgroup: Binary Tetrahedral, *Phys. Rev. D* **106**, 114501 (2022), [arXiv:2208.12309](https://arxiv.org/abs/2208.12309) [hep-lat].
- [49] G. T. Fleming, H. Lamm, K. Tame-Narvaez, and P. Vanderschuer, SWIFTbot: a pipeline for efficient qudit gate-set discovery, <https://github.com/petergerrit/quai> (2026).
- [50] A. Sawicki and K. Karnas, Universality of Single-Qudit Gates, *Ann. Henri Poincaré* **18**, 3515 (2017), [arXiv:1610.00547](https://arxiv.org/abs/1610.00547) [quant-ph].
- [51] H. Kesten, Symmetric random walks on groups, *Trans. Amer. Math. Soc.* **92**, 336 (1959).
- [52] B. D. McKay, The expected eigenvalue distribution of a large regular graph, *Linear Algebra Appl.* **40**, 203 (1981).
- [53] A. Lubotzky, *Discrete Groups, Expanding Graphs and Invariant Measures* (Birkhäuser, 1994).
- [54] O. Parzanchevski and P. Sarnak, Super-golden-gates for $PU(2)$, *Adv. Math.* **327**, 869 (2018), [arXiv:1704.02106](https://arxiv.org/abs/1704.02106) [math.NT].
- [55] M. S. Alam, S. Hadfield, H. Lamm, and A. C. Y. Li (SQMS), Primitive quantum gates for dihedral gauge theories, *Phys. Rev. D* **105**, 114501 (2022), [arXiv:2108.13305](https://arxiv.org/abs/2108.13305) [hep-lat].
- [56] M. B. Hastings and J. Haah, Distillation with sublogarithmic overhead (2018), [arXiv:1709.03543](https://arxiv.org/abs/1709.03543) [quant-ph].
- [57] D. Litinski, Magic state distillation: Not as costly as you think, *Quantum* **3**, 205 (2019), [arXiv:1905.06903](https://arxiv.org/abs/1905.06903) [quant-ph].
- [58] V. V. Albert and P. Faist, The Error Correction Zoo, <https://errorcorrectionzoo.org> (2024).
- [59] S. Forest, D. Gosset, V. Kliuchnikov, and D. McKinnon, Exact synthesis of single-qubit unitaries over Clifford-cyclotomic gate sets, *J. Math. Phys.* **56**, 082201 (2015), [arXiv:1501.04944](https://arxiv.org/abs/1501.04944) [quant-ph].
- [60] G. Duclos-Cianci and D. Poulin, Reducing the quantum-computing overhead with complex gate distillation, *Phys. Rev. A* **91**, 042315 (2015), [arXiv:1403.5280](https://arxiv.org/abs/1403.5280) [quant-ph].
- [61] E. T. Campbell and J. O’Gorman, An efficient magic state approach to small angle rotations, *Quantum Sci. Technol.* **1**, 015007 (2016), [arXiv:1603.04230](https://arxiv.org/abs/1603.04230) [quant-ph].
- [62] E. T. Campbell and M. Howard, Magic state parity-checker with pre-distilled components, *Quantum* **2**, 56 (2018), [arXiv:1709.02214](https://arxiv.org/abs/1709.02214) [quant-ph].
- [63] E. Kubicsha and I. Teixeira, Quantum Codes from Twisted Unitary t -groups, *Phys. Rev. Lett.* **133**, 030602 (2024), [arXiv:2402.01638](https://arxiv.org/abs/2402.01638) [quant-ph].
- [64] A. Krishna and J.-P. Tillich, Towards low overhead magic state distillation, *Phys. Rev. Lett.* **123**, 070507 (2019), [arXiv:1811.08461](https://arxiv.org/abs/1811.08461) [quant-ph].
- [65] L. Golowich and V. Guruswami, Asymptotically Good Quantum Codes with Transversal Non-Clifford Gates, in *57th Annual ACM Symposium on Theory of Computing (STOC ’24)* (2024) [arXiv:2408.09254](https://arxiv.org/abs/2408.09254) [quant-ph].
- [66] Q. T. Nguyen, Good binary quantum codes with transversal CCZ gate, in *57th Annual ACM Symposium on Theory of Computing (STOC ’25)* (2025) pp. 697–706.
- [67] S. Prakash, A. Jain, B. Kapur, and S. Seth, Normal form for single-qutrit Clifford+ T operators and synthesis of single-qutrit gates, *Phys. Rev. A* **98**, 032304 (2018), [arXiv:1803.03228](https://arxiv.org/abs/1803.03228) [quant-ph].
- [68] A. N. Glaudell, N. J. Ross, and J. M. Taylor, Canonical forms for single-qutrit Clifford+ T operators, *Annals Phys.* **406**, 54 (2019), [arXiv:1803.05047](https://arxiv.org/abs/1803.05047) [quant-ph].
- [69] A. R. Kalra, M. Mosca, and P. Valluri, Synthesis and Arithmetic of Single Qutrit Circuits, *Quantum* **9**, 1647 (2025), [arXiv:2311.08696](https://arxiv.org/abs/2311.08696) [quant-ph].